

ANDREW TRUONG

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SUMMARY

Entry-level hardware engineer with hands-on experience in bare-metal firmware development, analog circuit design, and PCB fabrication gained through academic projects and research. Designed and validated a full STM32-based lab instrument and a PPG biosensor signal chain from schematic to working prototype. Contributed to two faculty research programs, including AI/ML research in Tokyo. Seeking a hardware or firmware engineering role in embedded systems, consumer electronics, or aerospace/defense.

EDUCATION

University of Oklahoma

Bachelor of Science, Computer Engineering

Aug 2022 - Dec 2025

Norman, OK

- **GPA:** 3.6/4.0
- **Achievements:** Dean's List: 6 semesters (2022, 2023, 2024)
- **Coursework:** Digital Design Lab, Analog Electronics, Computer Architecture, Embedded Systems, Digital Signal Processing

WORK EXPERIENCE

University of Oklahoma

Undergraduate Research Assistant

Aug 2025 - Dec 2025

Norman, Oklahoma

- Conducted foundational research for a faculty-led smartwatch health monitoring system; findings directly contributed to professor's hardware platform selection and prototype development
- Surveyed on-device ML architectures (HiDRA, IoTCloak, OmniBuds) and benchmarked ECG, PPG, SpO2, and sleep staging accuracy across four platforms, producing a comparative analysis that narrowed the hardware search space for the research system

University of Electro-Communications

Machine Learning Research Assistant

Apr 2025 - Aug 2025

Tokyo, Japan

- Built Python data pipelines processing 4,000+ text datasets for NLP research, accelerating data preprocessing and enabling faculty to focus on model development
- Delivered bi-weekly technical presentations to faculty and research staff, directly contributing to an active research publication pipeline

Paycom

IT Systems Engineering Intern

May 2024 - Aug 2024

Oklahoma City, OK

- Built telemetry pipeline with Splunk, Prometheus, and Grafana aggregating metrics from 500+ VMs; developed automated VMware performance monitoring that reduced incident response time for the infrastructure team

Spiers New Technologies

Engineering Intern

Aug 2021 - May 2022

Oklahoma City, OK

- Designed custom camera mounting brackets in Autodesk Inventor and configured Axis camera network with PoE switches, reducing installation time and providing continuous video coverage across the facility

HARDWARE PROJECTS

Portable Digital Lab Tool

University of Oklahoma

Aug 2025 - Dec 2025

- Designed 3-in-1 lab instrument (power supply, ohmmeter, function generator) using STM32G031 with bare-metal C firmware (zero external libraries)
- Developed digitally-controlled DC power supply (0-3.3V, 0.05V resolution) and programmable function generator (10 Hz - 50 kHz) using PWM generation with RC filtering and hardware timer configuration
- Designed I2C peripheral system controlling dual MCP23017 I/O expanders driving four 7-segment displays and 4x4 matrix keypad, reducing interface from 32 pins to 2 wires
- Debugged MCP23017 current limiting issue after burning out four displays, resolving with 150Ω resistors and validating signal integrity with oscilloscope

Optical Heart Rate Monitor (PPG Biosensor)

University of Oklahoma

Aug 2024 - Dec 2024

- Designed photoplethysmography sensor using infrared LED and photodiode with 3-stage analog signal chain: transimpedance amplifier, active low-pass filter ($f_c \approx 2.34$ Hz), and secondary amplifier (5.55 V/V gain)
- Implemented RC integration circuit ($\tau = 68$ ms) for frequency-to-voltage conversion and comparator network for automatic heart rate zone classification (60, 120, 180 BPM) with LED indicators
- Conducted comprehensive circuit simulation in Multisim achieving 99.78% annual reliability and validated accuracy against commercial fitness trackers
- Mitigated motion artifacts and ambient light interference through analog filtering and optical shielding

TECHNICAL SKILLS

- **Embedded Systems:** STM32 microcontrollers, bare-metal C programming, I2C, SPI, PWM, ADC (12-bit), hardware timers, GPIO, UART, real-time firmware, interrupt-driven systems, register-level programming
- **Hardware Design:** Analog circuit design (op-amps, active filters, comparators, transimpedance amplifiers), PCB design (Eagle CAD, 2-layer boards), biosensor development, signal conditioning, hardware debugging (oscilloscopes, logic analyzers, multimeters)
- **Programming:** C (embedded/bare-metal), C++, Python (NumPy, pandas, scikit-learn), Java, Git/GitHub, MATLAB

AWARDS & HONORS

- Dean's List - University of Oklahoma (6 semesters)
- Eagle Scout
- 3rd Place - National Make a Medical Device Competition